

Exercise 4.1 (Laplace transform and its inverse Laplace transform)

$$f(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases}$$

Find the Laplace transform

$$F(s) = \int_0^\infty f(t) e^{-st} dt = \int_0^1 1 \cdot e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^1 = -\frac{1}{s} e^{-s} + \frac{1}{s} = \frac{1 - e^{-s}}{s}$$

Exercise 4.2 (Inverse Laplace transform) Find the inverse Laplace transform of the function $F(s) = \frac{1 - e^{-s}}{s}$ and sketch the graph of the function $f(t)$.

$$\begin{aligned} F(s) &= \frac{1}{s} - \frac{e^{-s}}{s} \\ f(t) &= 1 - 1 \cdot \delta(t-1) \\ f(t) &= \begin{cases} 1 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases} \end{aligned}$$

Exercise 4.3

$$\begin{aligned} f(t) &= \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases} \\ F(s) &= \int_0^\infty f(t) e^{-st} dt = \int_0^1 1 \cdot e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^1 = -\frac{1}{s} e^{-s} + \frac{1}{s} = \frac{1 - e^{-s}}{s} \\ f(t) &= \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases} \end{aligned}$$

Exercise 4.4 (Laplace transform and its inverse Laplace transform)

Exercise 4.5 (Laplace transform and its inverse Laplace transform) Find the Laplace transform of the function $f(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases}$ and sketch the graph of the function $f(t)$.

$$F(s) = \int_0^\infty f(t) e^{-st} dt = \int_0^1 1 \cdot e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^1 = -\frac{1}{s} e^{-s} + \frac{1}{s} = \frac{1 - e^{-s}}{s}$$

Exercise 4.6 (Laplace transform and its inverse Laplace transform)

Exercise 4.7 (Laplace transform and its inverse Laplace transform) Find the Laplace transform of the function $f(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases}$ and sketch the graph of the function $f(t)$.

Exercise 4.8 (Laplace transform and its inverse Laplace transform)

$$F(s) = \int_0^\infty f(t) e^{-st} dt = \int_0^1 1 \cdot e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^1 = -\frac{1}{s} e^{-s} + \frac{1}{s} = \frac{1 - e^{-s}}{s}$$

Exercise 4.9 (Laplace transform and its inverse Laplace transform)

$$F(s) = \int_0^\infty f(t) e^{-st} dt = \int_0^1 1 \cdot e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^1 = -\frac{1}{s} e^{-s} + \frac{1}{s} = \frac{1 - e^{-s}}{s}$$

Abstract: A review of the literature on the effects of the environment on the development of the child is presented. The review is organized into three sections: (1) the environment and the child's development; (2) the environment and the child's behavior; and (3) the environment and the child's health. The review is based on a search of the literature in the field of environmental health and development. The review is organized into three sections: (1) the environment and the child's development; (2) the environment and the child's behavior; and (3) the environment and the child's health. The review is based on a search of the literature in the field of environmental health and development.

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Die in (1) definierten Funktionen α und β sind die einzigen Funktionen, die die folgenden Eigenschaften besitzen:

- (i) α ist eine Abbildung von $\mathbb{R}^n$ in $\mathbb{R}^n$, β ist eine Abbildung von $\mathbb{R}^n$ in $\mathbb{R}^n$.
- (ii) α ist eine bijektive Abbildung, β ist eine bijektive Abbildung.
- (iii) α ist eine lineare Abbildung, β ist eine lineare Abbildung.
- (iv) α ist eine symmetrische Abbildung, β ist eine symmetrische Abbildung.
- (v) α ist eine Abbildung, die die Eigenschaft $\alpha(x) = x$ für alle $x \in \mathbb{R}^n$ erfüllt.
- (vi) β ist eine Abbildung, die die Eigenschaft $\beta(x) = x$ für alle $x \in \mathbb{R}^n$ erfüllt.

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